Project: Design and Development of a Low-Cost, Self-Righting Robotic Platform for Confined Space Inspection

Problem: Robotic solutions for inspecting hazardous or hard-to-reach environments (e.g., Boston Dynamics' Spot) are considerably expensive, which results in a high barrier to entry for most individuals or businesses. Additionally alternatives such as Unitree's variations of the "robot dog," whilst cheaper, suffer from the same preexisting issue of size. Its boxy structure limits its operation to areas in which human access is most likely available. There is a clear need for a smaller, more affordable alternative that can operate as a robust robotic platform for basic remote visual assessment.

Project Aim & Objectives The primary aim is to design and build a functional proof-of-concept of a low-cost, bi-hemispherical rolling robot capable of effective navigation and inspection in confined spaces.

Key Objectives:

- Design a mechanically robust and environmentally sealed chassis with a dual motor differential-drive system.
- Develop and implement an active self-righting mechanism using a simple counterweight or electronically activated system.
- Integrate a modular sensor payload featuring a live-feed video camera for remote operation.
- Establish a wireless control system for manual piloting.
- Evaluate the prototype's mobility and inspection capabilities in a simulated environment.

3. Scope

- **In-Scope:** Mechanical design (CAD), electronics integration (microcontroller, motors, sensors), firmware development for control and self-righting, and construction of a physical prototype.
- **Out-of-Scope:** Advanced autonomous navigation (SLAM), object manipulation, and software for data analysis. The focus is on a robust teleoperated platform.

4. Proposed Solution The proposed solution is a pill-shaped robot featuring two large, treaded hemispheres for movement. A central, sealed cylindrical body will house all core components: battery, motors, control electronics, and the sensor package. The key technical innovations will be the miniaturised design and the self-righting system, enabling the robot to recover if it is knocked onto its side, thus ensuring mission reliability.

5. Significance & Impact This project aims to deliver a practical and affordable alternative to high-end inspection robots. By creating a durable, simple, and effective platform, it has the potential to enhance workplace safety across construction, industrial maintenance, and emergency response sectors by minimising human exposure to hazardous environments.